

W1 SETS NUMBERS COMPLEX

Sets: $A \subset B \iff A \cap A = A \in B; A \cup B, A \cap B, A^c$; De Morgan $(A \cup B)^c = A^c \cap B^c$.

Logic: converse need not hold; prove equality by two inclusions; counterexample disproves universal stmt.

Intervals: open endpoints not included; natural domain from all restrictions; denom $\neq 0$, even root radicand ≥ 0 , log arg > 0 .

Complex: $z = a + bi, \bar{z} = a - bi, |z| = \sqrt{a^2 + b^2}, z\bar{z} = |z|^2, 1/z = \bar{z}/|z|^2$.

Polar: $z = re^{i\theta} = r(\cos \theta + i \sin \theta), r = |z|, \arg z = \theta + 2k\pi$.

Powers: de Moivre $(re^{i\theta})^n = r^n e^{in\theta}$; roots of $z^n = Re^{i\phi}$: $R^{1/n} e^{i(\phi+2k\pi)/n}$.

Pitfalls: quadrant sets use signs of $\Re z, \Im z$; real polynomial nonreal roots occur in conjugate pairs.

W2 FUNCTIONS COMPLEX

Function: domain/codomain part of data; image/range may be smaller than codomain.

Injective: $f(x_1) = f(x_2) \implies x_1 = x_2$ or horizontal line test.

Surjective: every codomain value hit; bijective: both, inverse exists.

Strict monotone on interval $\rightarrow$ injective; derivative > 0 gives increasing, but derivative > 0 needs care.

Inverse: $f^{-1}(f(x)) = x$ on domain; $f(f^{-1}(y)) = y$ on range; restrict domain if needed.

Complex division: multiply by conjugate; argument principal usually $(-\pi, \pi]$.

form	data	use
cart	$a + bi$	add/subtract
polar	$re^{i\theta}$	powers/roots
exp	$e^{i\theta}$	rotation

Exam signal: $Z \rightarrow \bar{Z}, n \mapsto 2n$ injective not surjective; codomain matters.

W3 LIMITS VECTORS

Limit laws: valid when component limits exist; indeterminate $0/0, \infty/\infty, 1^\infty$ need transform.

$\sin x \sim x, 1 - \cos x \sim x^2/2, e^x = 1 + x + x^2/2 + \dots, \ln(1+x) = x - x^2/2 + \dots$.

Series limit: expand to first nonzero surviving term; do not cancel only leading terms when numerator has higher-order cancellation.

Vectors: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$; $\mathbf{a} \times \mathbf{b}$ perpendicular, area $= |\mathbf{a} \times \mathbf{b}|$.

$\text{proj}_{\mathbf{v}}$ $= \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$; distance point-plane $= |ax_0 + by_0 + cz_0 - d|/\sqrt{a^2 + b^2 + c^2}$.

Pitfall: dot product tests angle/projection; cross product gives area/normal.

W4 CONT DIFF VECTORS

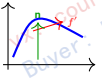
Continuity: limit=value; piecewise join needs left value=right value, differentiability also left deriv=right deriv.

Derivative: product, quotient, chain, implicit. If $F(x, y) = 0$, $y' = -F_x/F_y$ when $F_y \neq 0$.

Critical point: $F'_x = 0$ or F'_y DNE inside domain; classify by sign chart or f'' test only when valid.

Concavity: $f'' > 0$ up, $f'' < 0$ down; inflection requires concavity change.

Lines/planes: line $\mathbf{r} = \mathbf{p} + t\mathbf{v}$; plane $\mathbf{n} \cdot (\mathbf{r} - \mathbf{p}) = 0$; three pts $\implies \mathbf{n} = (Q - P) \times (R - P)$.



W5 LIMITS VECTORS II

Asymptotics: horizontal use leading terms or limits; vertical where denom zero and one-sided blow-up; oblique via division.

$\lim_{x \rightarrow \infty} (1 + a/x)^{bx} = e^{ab}$; take logs for powers $u(x)^{v(x)}$.

Vector identities: $(A + B) \times (cA + dB) = (d - c)A \times B$; anti-commute $B \times A = -A \times B$.

Line-plane intersection: substitute param line into plane; no solution parallel/noncontained, all t contained.

Skew lines: in $\mathbb{R}^3$, not parallel and no intersection; solve vector equation for two params.

Recall focus: projection, parallelogram area, plane through points, tangent/limit traps.

W6 OPTIMISATION PLANES

Optimisation: identify domain incl endpoints/asymptotes; solve critical pts; compare values or sign changes.

Closed interval: max/min among endpoints and critical pts. Open/unbounded: check limits; extrema may not exist.

Second derivative: $f''(c) > 0$ min, < 0 max, $= 0$ inconclusive.

Geometry: plane normal from coefficients; vector parallel to plane iff dot normal $= 0$; distance between parallel planes $|d_1 - d_2|/|\mathbf{n}|$.

Sketch: domain $\rightarrow$ intercepts $\rightarrow$ asymptotes $\rightarrow$ monotonicity $\rightarrow$ concavity $\rightarrow$ labelled graph.

Pitfall: a vertical asymptote may split domain; do not classify extrema across missing point.

W7 OPTIMISATION SYSTEMS

Linear system: coefficient matrix, augmented matrix; RREF tells none/one/infinately many.

Unique solution iff pivot in every variable column iff square A invertible iff $\det A \neq 0$.

Parameter systems: row reduce symbolically; branch when pivot expression $= 0$; then test consistency.

Matrix inverse: A^{-1} exists iff nonsingular; solve $Ax = b$ by row reduction, inverse, or LU-style elimination.

Rows: swap changes det sign; scale row by c scales det by c ; replacement unchanged.

Exam signal: if row ops turn A into I , track determinant operation factors backward.

W8 TAYLOR MATRICES

$f(x) = \sum f^{(n)}(a)(x-a)^n/n!$; $R_n = f^{(n+1)}(\xi)(x-a)^{n+1}/(n+1)!$.

W9 RIEMANN MATRICES

Riemann integral: $\int_a^b f = \lim \sum f(x_i^*) \Delta x_i$; area signed; monotone/continuous integrable.

FTC: if $G(x) = \int_a^x f(t) dt$, then $G'(x) = f(x)$; variable bounds: $\frac{d}{dx} \int_u(x)^v(x) f(t) dt = f(v)v' - f(u)u'$.

Riemann sum: convert $\sum_{k=1}^n f(k/n) \frac{1}{n} \rightarrow \int_0^1 f(x) dx$; beware extra n factors.

Matrix powers: exploit cycles; triangular/nilpotent parts, or diagonalisation later.

Rank: pivots count; nullity $= n - \text{rank } A$; homogeneous system always consistent.

W10 INTEGRATION DETERMINANTS

Methods: substitution, parts, partial fractions, trig identities, improper tests.

$\int u dv = uv - \int v du$; for $u = \sqrt{t}, dt = 2u du$ and change limits.

Improper: infinite interval or vertical singularity; define as limit; p -test $\int_1^\infty x^{-p} dx$ conv iff $p > 1, \int_0^1 x^{-p} dx$ conv iff $p < 1$.

Determinants: area/volume scale; cofactor expansion; triangular det product diagonal.

Singularity: $\det A = 0 \iff$ rows/cols dependent $\iff$ no inverse.

Recall focus: determinant row operations and improper integral convergence are high-frequency MCQ traps.

W11 INT EIGENVALUES

Eigen: $A\mathbf{v} = \lambda\mathbf{v}, \mathbf{v} \neq 0$; solve $(A - \lambda I)\mathbf{v} = 0$; characteristic polynomial $\det(A - \lambda I)$.

For triangular A , eigenvalues are diagonal entries; $\det A = \prod \lambda_i, \text{tr } A = \sum \lambda_i$, counting alg mult.

Integration apps: area between curves f (top-bottom) dx ; volume washers $\pi \int (R^2 - r^2) dx$; shells $2\pi \int \text{radius} \cdot \text{height} dx$.

Eigenvectors: eigenspaces for distinct eigenvalues linearly independent. If $x = \sum c_i v_i, A^k x = \sum c_i \lambda_i^k v_i$.

Pitfall: repeated eigenvalue not automatically enough eigenvectors.

W12 APP INT EIGENVECTORS

Rotation choice: around x -axis often washers in x ; around y -axis often shells in x unless inverse functions.

Curve length $L = \int_a^b \sqrt{1 + (y')^2} dx$; average value $= \frac{1}{b-a} \int_a^b f$.

Diagonalisation: $A = PDP^{-1}$ iff basis of eigenvectors; then $A^n = PD^n P^{-1}$.

Algebraic vs geometric: alg mult root multiplicity; geom mult eigenspace dimension. Diagonalizable iff sum geom mults $= n$.

Complex eigenvalues: real matrices may have nonreal conjugate pairs; over $\mathbb{C}$ more diagonalisation possible.

Exam signal: $P^{-1} A^k P = D^k$ when $A = PDP^{-1}$, not $PD^k P^{-1}$ unless the similarity basis.

W13 APP INT DIAGONALISATION

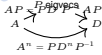
Diagonalisation workflow: find eigenspaces $\rightarrow$ bases for eigenspaces $\rightarrow$ assemble P columns $\rightarrow D$ same order $\rightarrow$ verify $AP = PD$.

If A has n distinct eigenvalues, diagonalizable. If one eigenvalue has alg mult m , need geom mult m .

Applications: recurrence/state transition $x_k = A^k x_0$; diagonalisation turns coupled system into scalar powers.

Non-diagonalizable: Jordan-like repeated eigenvalue with too few eigenvectors; e.g. $(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix})$.

Final triage: MCQ first pass: domain/codomain, determinant scaling, eigen multiplicity, Taylor signs; written: show working, state domain, label graph/asymptotes.



Curve sketch:

domain; intercepts; asymptotes; f' sign; f'' sign; endpoints/one-sided limits; labelled critical/inf pts.

CALCULUS PATTERNS

Taylor use cases: approximate values, compute hard limits, sign of error. If terms cancel, expand one order further than expected.

f	f'	f''	sign
increasing	+	-	
decreasing	-	-	
concave up	+	+	
concave down	-	+	
local max	$\rightarrow$	often $-$	
local min	$\rightarrow$	often $+$	

$\ln(1+u) = u - u^2/2 + u^3/3 - u^4/4 + \dots$; substitute whole u first, then collect powers.

$(1+u)^\alpha = 1 + \alpha u + \alpha(\alpha-1)u^2/2 + \alpha(\alpha-1)(\alpha-2)u^3/6 + \dots$.

Remainder sign: for Taylor degree n , $R_n(x) = f^{(n+1)}(\xi)(x-a)^{n+1}/(n+1)!$; check sign of derivative and displacement.

FTC variable bounds: $H(x) = \int_{h(x)}^{f(x)} F(t) dt \rightarrow H' = F(b) b' - F(a) a'$.

Substitution: choose inside derivative; change bounds for definite integrals; after $u = \sqrt{x}, t = u^2, dt = 2u du$.

Paris: choose u to simplify under derivative; LIATE guide: log, inverse trig, algebraic, trig, exp.

Partial fractions: factor denom; repeated linear uses $A/(x-a) + B/(x-a)^2$; irreducible quadratic uses linear numerator.

Improper workflow: split at every singularity; replace bad endpoint by limit; convergence requires every split limit finite.

integral	conv	when value
$\int_1^\infty x^{-p}$	$p > 1$	$1/(p-1)$
$\int_0^1 x^{-p}$	$p < 1$	$1/(1-p)$
$\int_1^\infty 1/x$	never	div

VECTOR GEOMETRY

Projection decomposition: $\mathbf{v} = \text{proj}_{\mathbf{u}} \mathbf{v} + \mathbf{v}_\perp$; verify by dotting $\mathbf{v}_\perp$ with $\mathbf{u}$.

$|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

Plane through 3 pts: $\mathbf{u} = Q - P, \mathbf{v} = R - P, \mathbf{n} = \mathbf{u} \times \mathbf{v}$, then $\mathbf{n} \cdot (x, y, z - P) = 0$.

Line relations: parallel iff direction vectors scalar multiples; intersect iff vector equation has consistent params; otherwise skew in $\mathbb{R}^3$.

Line-plane: denom $\mathbf{n} \cdot \mathbf{v}$ if nonzero one intersection; if zero, either parallel/disjoint or contained.

task	tool
angle vectors	dot product
area parallelogram	cross product
normal to plane	cross of directions
distance to plane	point-plane formula
projection	dot ratio

Common trap: vector parallel to a plane is perpendicular to the normal, not equal to the normal.

MATRIX SYSTEMS

RREF interpretation: pivot every var $\implies$ unique if consistent; free var $\implies$ infinitely many if consistent; row $0 = \text{nonzero}$ impossible.

$Ax = b$; if A square and $\det A \neq 0$, exactly one solution for every b ; if $\det A = 0$, need rank check.

Inverse by row reduction: augment $[A|I] \rightarrow [I|A^{-1}]$; if left cannot reach I , no inverse.

row op	effect on det	note
swap	$\times(-1)$	
scale row by c	$\times c$	if $c=0$
replace $R_i + cR_j$	same	area/vol unchanged

Det identities: $\det A^T = \det A$; $\det(A^k) = (\det A)^k$; $\det(P^{-1} A P) = \det A$; similar matrices share char poly.

Triangular: determinant and eigenvalues from diagonal; inverse exists iff all diagonal entries nonzero.

Dimension sanity: $A_{m \times n} B_{p \times q}$ exists iff $n = p$, result $m \times q$; $A^T M A$ needs M sized $m \times m$ if A is $m \times n$.

EIGEN DIAGNOSTICS

Characteristic polynomial: convention may be $\det(A - \lambda I)$ or $\det(\lambda I - A)$; roots same.

Eigenspace $E_\lambda = \text{null}(A - \lambda I)$; eigenvector cannot be 0 .

Diagonalizable tests: enough independent eigenvectors; n distinct eigenvalues sufficient; repeated eigenvalue requires geom mult=alg mult.

situation	diag?	why
distinct n eivals	yes	indep eigvecs
repeated, full eigspace	yes	basis exists
Jordan block	no	too few eigvecs
symmetric real	yes	orthogonal eigbasis

P/D order: if $P = [v_1 \ v_2 \dots]$, then $D = \text{diag}(\lambda_1, \lambda_2, \dots)$ in same order. Verify $AP = PD$.

$P^{-1} A^k P = D^k$ and $A^k x = \sum c_i \lambda_i^k v_i$ when $x = \sum c_i v_i$.

Trap: $\lambda = 0$ allowed as eigenvalue; it means singular matrix. Zero vector is not an eigenvector.

MINI EXAMPLES

Function: $f: Z \rightarrow Z, f(n) = 2n$ inj since $2m = 2n \implies m = n$, not sur since odd outputs missed.

Radical limit: $\frac{\sqrt{x+2} - 2}{x} \cdot \frac{\sqrt{x+2} + 2}{\sqrt{x+2} + 2} \rightarrow \frac{1}{\sqrt{x+2} + 2}$.

Taylor limit: $\sin(3x) - 3 \sin x = (3x - \frac{27x^3}{6}) - (3x - \frac{3x^3}{6}) + \dots = -4x^3 + \dots$.

Projection: $\text{proj}_{(1,2,3)}(3,2,1) = \frac{19}{14}(1,2,3)$.

Plane: pts $(1,0,2), (3,1,1), (2,-1,3)$ give dirs $(2,-1,-1), (1,-1,1)$, normal $(0,-3,-3)$, plane $y + z = 2$.

Det row ops: if $A \rightarrow I$ by swap then row scale by 3, then $1 = (-3) \det A$, so $\det A = -1/3$.

Eigen:

$A = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$ has char $\lambda^2 + \lambda - 6$, eivals $2, -3$, eigvecs $(3,1), (-2,1)$.

EXAM ERROR CHECKS

Before final answer: state natural domain; check endpoints; units/axis of rotation; whether area is signed or absolute; whether codomain affects surjectivity.

MCQ elimination: test dimensions; test special value $0, 1, -1$; check sign/quadrant; repeated eigenvalue is usually a trap; determinant scaling by c^n not c .

Written answers: for sketch, label asymptote + extrema + concavity; for systems, show row operations; for eigen, show $(A - \lambda I)\mathbf{v} = 0$; for integrals, show changed bounds.

No calculators style: keep exact forms; simplify only when safe; avoid decimal mess question asks.

If stuck: convert geometry to vectors, function behaviour to sign charts, matrix question to pivots/determinants/eigenspaces, integral app to a labelled slice.

HIGH-FREQ MCQ TRAPS

Injective/surjective: $x^3 - 3x: \mathbb{R} \rightarrow \mathbb{R}$ is surj not inj; $x^4 + x^2: \mathbb{R} \rightarrow \mathbb{R}$ neither; $e^x: \mathbb{R} \rightarrow (0, \infty)$ bijective.

Codomain trap: $2n: Z \rightarrow Z$ misses odds; same formula $Z \rightarrow 2Z$ is bijective.

Taylor coefficient: $\ln(1+2x) = 2x - 2x^2 + \frac{x^3}{3} - 4x^4 + \dots$; common wrong term $\frac{x^3}{3}$.

Matrix power cycle: $M = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ has $M^3 = I$, so $M^{100} = M$.

Nilpotent shift: $A = \lambda I + N, N^2 = 0$ for 2×2 Jordan block, so $A^n = \lambda^n I + n \lambda^{n-1} N$.

Det scaling: for 4×4 , $\det(2A^{-1} A^T) = 2^4 \det(A^{-1}) \det(A^T) = 16$.

Eigen action: if $x = 2v_1 - 3v_2 + v_3, A^2 x = 2\lambda_1^2 v_1 - 3\lambda_2^2 v_2 + \lambda_3^2 v_3$.

Implicit derivative: $e^y + y = x+1: (e^y + 1)y' = 1$, then $e^y(y')^2 + (e^y + 1)y'' = 0$.

Quadrant product: QI,QIII $\Re z \Im z > 0$; QII,QIV $\Re z \Im z < 0$; axes excluded if strict.

Root construct: if real coeff polynomial has root $1+i$, also $1-i$; product factor $z^2 - 2z + 2$.

WRITTEN Q TECHNIQUES

Graph analysis answer: natural domain; g' zeros/DNE; classify; limits at asymptotes/infinity; g'' sign; sketch w labels.

Taylor written: state n th series; substitute; polynomial degree; remainder formula/sign/bound; final approximation with interval condition.

Plane/vector written: compute direction vectors; cross for normal; write equation; simplify but keep sign consistent; check one point.

System written: write augmented matrix; row-reduce; branch parameter cases; translate RREF to solution set; verify if asked.

Integral application: draw/identify region; choose washers/shells/slices; set bounds; radius/height; evaluate; state exact volume/area.

Eigen written: char poly; eigenvalues; solve eigenspaces; if diagonalising, assemble P, D and state order, use $A = PDP^{-1}$.

TAYLOR DETAIL

$\sin x - x + x^3/6 = x^5/120 - x^7/5040 + \dots$, so divided by $x^5 \rightarrow 1/120$.

$e^{-x^2} = 1 - x^2 + x^4/2 - x^6/6 + \dots$; degree 4 polynomial $1 - x^2 + x^4/2$.

$\ln(\cos x) = -x^2/2 - x^4/12 - x^6/45 + \dots$; derive via $\ln(1+u)$ with $u = \cos x - 1$.

Bound idea: if $|f^{(n+1)}(t)| \leq M$ between a, x , then $|R_n(x)| \leq M|x-a|^{n+1}/(n+1)!$.

Alternating series estimate: if alternating decreasing terms, error after truncation no larger than first omitted term and has sign of first omitted term.

INTERGAL DETAIL

Washer vs shell: rotate around horizontal axis with vertical slices $\implies$ washers; around vertical axis with vertical slices $\implies$ shells.

Between $y = x$ and $y = x^2$ on $[0, 1]$: area $= \int_0^1 (x - x^2) dx = 1/6$;

about y -axis shells $V = 2\pi \int_0^1 x(x - x^2) dx = \pi/6$.

Between $y = x^2$ and $y = x^3$ on $[0, 2]$: washers about x -axis $V = \pi \int_0^2 ((2x)^2 - (x^2)^2) dx = 64\pi/15$.

Average value: for x^2 on $[0, 3]$, avg $= \frac{1}{3} \int_0^3 x^2 dx = 3$.

Curve length: for $y = \ln(\sin x), \sqrt{1 + \cot^2 x} = \csc x$, integrate $\int \csc x dx = \ln|\tan(x/2)| + C$.

LINEAR ALG DETAIL

2x2 eigens: $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, char $\lambda^2 - (a+d)\lambda + (ad-bc)$.

λ alg mult 2 in 2×2 : diagonalizable iff $A = \lambda I$ or eigenspace dimension 2 ; otherwise one eigendirection only.

Rank cases 3 unknowns: rank 3 unique; rank 2 consistent line of solutions; rank 1 consistent plane of solutions; inconsistent if augmented rank bigger.

Area/volume transform: determinant absolute value is scale factor; sign is orientation. Columns as parallelogram/parallelepiped edges.

Similarity: A and D have same determinant, trace, eigenvalues; diagonalisation changes coordinates, not the linear map's intrinsic spectrum.

FAST SELF-TEST

Can I answer? If a question asks "must be true", look for counterexample; if "which statement false", test edge case repeated eigenvalue/domain endpoint.

Check signs: c vs d curve division sign; determinant row swap sign; shell height top-bottom; improper integral positive but may diverge.

Check dimensions: vector vs scalar; matrix product size; $P^{-1} A P$ only if P square invertible; cross product only in $\mathbb{R}^3$ in this course.

Check endpoints: closed interval optimisation, area, cone intersections, integral bounds after substitution, piecewise continuity join point.

Check definitions: local vs absolute extremum; image vs codomain;